

SABEE GREWAL

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Links: [Personal Website](#), [Google Scholar Profile](#).

EDUCATION

IN PROGRESS **Ph.D. in Computer Science** *University of Texas at Austin*.
Advisor: Scott Aaronson

2021 **M.S. in Electrical and Computer Engineering** *University of Texas at Austin*.

2019 **B.S. in Physics** *University of Texas at Austin*.

2017 **B.S. in Electrical and Computer Engineering** *University of Texas at Austin*.

PUBLICATIONS

Author ordering in my field is alphabetical by default (all exceptions below are evident).

- S. Grewal, V. Iyer, W. Kretschmer, D. Liang. Agnostic Tomography of Stabilizer Product States. *In Submission*. [\[ARXIV\]](#)
- S. Grewal, V. Iyer, W. Kretschmer, D. Liang. Pseudoentanglement Ain't Cheap. *In Submission*. [\[ARXIV\]](#)
- S. Aaronson, S. Grewal, V. Iyer, S. C. Marshall, R. Ramachandran. PDQMA = DQMA = NEXP: QMA With Hidden Variables and Non-collapsing Measurements. *In Submission*. [\[ARXIV\]](#)
- S. Grewal, J. Yirka. The Entangled Quantum Polynomial Hierarchy Collapses. *In Submission*. [\[ARXIV\]](#) [\[ECCC\]](#)
- S. Grewal, V. Iyer, W. Kretschmer, D. Liang. Efficient Learning of Quantum States Prepared With Few Non-Clifford Gates. *Presented at QIP 2024. In Submission*. [\[ARXIV\]](#)
- S. Grewal, V. Iyer, W. Kretschmer, D. Liang. Improved Stabilizer Estimation via Bell Difference Sampling. *56th Annual ACM Symposium on Theory of Computing (STOC 2024). Presented at QIP 2024*. [\[ARXIV\]](#)
- S. Aaronson, S. Grewal. Efficient Tomography of Non-Interacting-Fermion States. *18th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2023), Leibniz International Proceedings in Informatics (LIPIcs)* 266, 12:1–12:18 (2023). [\[ARXIV\]](#) [\[DOI\]](#)
- S. Grewal, V. Iyer, W. Kretschmer, D. Liang. Low-Stabilizer-Complexity Quantum States Are Not Pseudorandom. *14th Innovations in Theoretical Computer Science Conference (ITCS 2023), Leibniz International Proceedings in Informatics (LIPIcs)* 251, 64:1–64:20 (2023). **ITCS 2023 Best Student Paper**. [\[ARXIV\]](#) [\[DOI\]](#)
- D. E. Koh, S. Grewal. Classical Shadows With Noise. *Quantum*: 6, 776 (2022). [\[ARXIV\]](#) [\[DOI\]](#)

EXPERIENCE

RESEARCH EXPERIENCE

SUMMER 2024 **Quantum Research Scientist Intern** IBM, Yorktown Heights, NY.

SPRING 2024 **Long-Term Visitor** Simons Institute for the Theory of Computing, Berkeley, CA.

2021–PRESENT **Graduate Research Assistant** Quantum Information Center, UT Austin.

SUMMER 2020 **Quantum Computing Research Intern** *Zapata Computing*, Boston, MA.
SUMMER 2019 **Quantum Computing Research Intern** *Zapata Computing*, Boston, MA.

ENGINEERING EXPERIENCE

FALL 2020 **Senior Software Engineer** *Zapata Computing*, Boston, MA.
2017–2019 **Software Engineer II** *Microsoft Corporation*, Redmond, WA.
SUMMER 2016 **Software Engineer Intern** *Microsoft Corporation*, Redmond, WA.
SUMMER 2015 **Software Engineer Intern** *Microsoft Corporation*, Redmond, WA.
SUMMER 2014 **CPU Design Intern** *Intel Corporation*, Austin, TX.

SELECTED TALKS & PRESENTATIONS

2024 **Learning Quantum States With Respect To The Stabilizer Formalsim.**
27th Conference on Quantum Information Processing (QIP 2024)
2023 **Efficient Tomography of Non-Interacting-Fermion States.**
18th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2023)
2023 **Low-Stabilizer-Complexity Quantum States Are Not Pseudorandom.**
14th Innovations in Theoretical Computer Science (ITCS 2023)

TEACHING

FALL 2023 **Teaching Assistant** *Quantum Information Science I*, UT Austin.
FALL 2021 **Head Teaching Assistant** *Introduction to Computing Systems*, UT Austin.
SPRING 2021 **Teaching Assistant** *Combinatorics and Graph Theory*, UT Austin.
FALL 2019 **Head Teaching Assistant** *Introduction to Computing Systems*, UT Austin.
SPRING 2019 **Teaching Assistant** *Software Testing*, UT Austin.
FALL 2016 **Teaching Assistant** *Engineering Programming Languages*, UT Austin.
FALL 2015 **Teaching Assistant** *Introduction to Computing Systems*, UT Austin.
FALL 2014 **Teaching Assistant** *Software Design and Implementation*, UT Austin.

SKILLS

Programming/Tools: Python, Java, Scala, C++, Bash, L^AT_EX, Git, AWS, Azure

SELECTED AWARDS

2023 **Best Student Paper Award** *14th Innovations in Theoretical Computer Science (ITCS 2023)* .
2020 **Teaching Excellence Award** *Cockrell School of Engineering*, UT Austin.
2015 **Teaching Excellence Award** *Cockrell School of Engineering*, UT Austin.
2014 **Teaching Excellence Award** *Cockrell School of Engineering*, UT Austin.
2014 **Distinguished College Scholar**, UT Austin, (for being in the top 4% of the college).

PROFESSIONAL SERVICE

Conference Reviewer: QIP 2023, QIP 2024, QCTIP 2024, TQC 2024, ICALP 2024.

Journal Review: *Theory of Computing Systems* (2024).

Last updated: April 2024